Quantitative Area Loss, Capacity Enclosures, and Mass-Defect Windows for Conformal Flow with Minimizing Frontiers

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Abstract

We examine the quantitative interfaces in the assigned manuscript on conformal flow with full, possibly singular minimizing frontiers. Under its finite-time density, positivity, and harmonic Hölder bounds, direct summation of swept volumes improves the displayed uniform-grid area-loss estimate from order $\epsilon^{\alpha/(2n)}$ to order $\epsilon^{\alpha/n}$. A nonuniform-grid version retains each individual swept volume. For an actual continuous flow, the coupled mass and capacity identities give a sharp two-endpoint upper bound for the instantaneous mass defect. Whole-exterior primal and equilibrated flux certificates give a second, independent defect bound without assigning classical normals to singular points. We derive the exact harmonic-flat tail form, exhibit radial normalizations, and identify the boundary and tail data needed to turn local Ram representations into valid certificates. These results do not construct the weak flow, prove singular detachment or a positive-mass theorem, or supply an effective global convergence time. Contribution assessment: C3/5, a source-specific improvement of the finite-time area-loss exponent and a sharp conditional mass-window certificate, with the global geometric inputs retained rather than certified.

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1 The Actual Geometric Problem

Put $n \geq 3$, $d = n-2$, $k = n-1$, $\omega = |S^{n-1}|$, and $p = 2k/d$. The source [1] concerns a connected asymptotically flat end exterior with its entire compact minimizing frontier included. The ambient metric extends smoothly through that frontier. In a fixed end chart,

$$g - \delta = O_2(r^{-\tau}), \quad \tau > d/2, \qquad m_{\mathrm{ADM}}(g) = \frac{1}{2k\omega} \lim_{R\to\infty} \int_{S_R} (\partial_j g_{ij} - \partial_i g_{jj}) \nu_\delta^i \, \mathrm{d}A_\delta. \tag{1}$$

The source also assumes $R_g \geq 0$, $R_g \in L^1$, and $R_g = O(r^{-n-\beta})$ for some $\beta > 0$. The filled side K is represented canonically, not by adding arbitrary measure-zero points. Its frontier is $\Sigma = \operatorname{supp} |D\mathbf{1}_K|$, and its full area is the reduced perimeter $A = \mathcal{H}_g^k(\partial^* K)$. All components and contact portions count. Using relative perimeter only in the open exterior would erase contact area. Local perimeter minimality on both sides is stronger than stationarity. The regular part is smooth and minimal; the imported singular-set bound is $\dim_{\mathrm{H}} \Sigma_{\mathrm{sing}} \leq n-8$.

The source's numerical conclusion is

$$m_{\mathrm{ADM}}(g) \geq \tfrac{1}{2}(A/\omega)^{d/k}. \tag{2}$$

We do not independently prove this global statement. Nor do we identify this as an asymptotically hyperbolic or spacetime constraint problem. There is no second fundamental form or Jang equation in the local results below.

For the harmonic-flat stage, $g_0 = H^{4/d}\delta$ near infinity, with $\Delta_\delta H = 0$ and $H \to 1$. A buried physical boundary B lies inside the initial filled region. The flow uses

$$g_t = u_t^{4/d} g_0, \qquad u_t = 1 + \int_0^t v_s \, \mathrm{d}s, \qquad \Delta_{g_0} v_t = 0 \text{ on } E_t = M \setminus K_t, \quad v_t|_{K_t} = 0, \quad v_t(\infty) = -e^{-t}. \tag{3}$$

The actual metric capacity potential is

$$w_t = 1 + v_t/u_t, \qquad c_t = \frac{1}{d\omega} \int_{E_t} |\nabla w_t|_{g_t}^2 \, \mathrm{d}V_{g_t}, \qquad \mu(t) = m(t) - c_t. \tag{4}$$

It is not the background potential $1 + e^t v_t$. Both equal one on the conductor and zero at infinity, but solve different metric problems. A test function may equal one on the buried collar; imposing zero Dirichlet data on B would change the problem. The equilibrium potential has finite homogeneous energy, but need not belong to L^2 when $n = 3, 4$.

1.1 Dependency Map and Retained Details

The following is a map of the source, not a certification of all its arrows.

1. The enclosure modules use full perimeter, compact containment, submodularity, and smooth-obstacle regularity. Strict negative inner mean curvature, with normal toward the end, gives detachment from a smooth obstacle.

2. The fully read `flow.tex` constructs discrete weighted hulls, proves uniform finite-time density and Hölder control, and proves discrete area preservation in the limit. Only afterwards does it prove nesting and freezing of the limiting conductors. This order must not be reversed.

3. The regularity/separation and weighted-minimizer modules address contact at singular points using regular cone patches, one common median normalization, and conformal forcing. A patchwise Hopf derivative alone is insufficient.

4. Mass-capacity comparison requires actual detached enclosures, smoothing, complete comparison metrics, and a positive-mass input. The fully read mass-evolution and estimates modules then obtain $m' = -2\mu$, $\mu \geq 0$, and the distribution inequality used in Section 3.

5. Completion, quantitative positive mass, the normalized end limit, and harmonic-flat approximation link these properties to (2). The source normalizes spatial end coordinates by $x = e^{-2t/d}y$; it does not drop the initial harmonic factor H from the end mass.

Our improvement concerns item 2, conditional on its already established finite-time inputs, and an exact quantitative consequence of item 4. The global obligations in items 1, 3, and 5 remain imported, including portions not literally inspected. No source mathematical correction is asserted.

2 Direct Swept-Volume Summation

Fix $T < \infty$. All distances, volumes, and Hölder seminorms in this section use the smooth reference metric g_0, not the evolving metric. Consider $0 = t_0 < t_1 < \cdots < t_N \leq T$, $h_j = t_j - t_{j-1}$, and nested compact canonical filled sets $K_0 \subset \cdots \subset K_N$. Write $\Delta V_j = \mathrm{Vol}_{g_0}(K_j \setminus K_{j-1})$ and $V = \sum_j \Delta V_j$. Assume the following actual-state bounds:

1. At each $x \in \partial K_j$ and $0 < r < r_*$, $\mathrm{Vol}_{g_0}(K_j \cap B_r(x)) \geq c_* r^n$, with $c_*, r_* > 0$ independent of j.

2. Continuous velocities $v_{j-1} \leq 0$ vanish on all of K_{j-1}, $|v_{j-1}| \leq B$, and $[v_{j-1}]_{C^{0,\alpha}} \leq L$ for one $0 < \alpha \leq 1$.

3. The update is $u_j = u_{j-1} + h_j v_{j-1}$, with $0 < \underline{u} \leq u_j \leq \overline{u}$. Define $A_j = \int_{\partial^* K_j} u_j^p \, \mathrm{d}\mathcal{H}^k_{g_0}$. Each previous grid hull permits K_j as a competitor, and the new hull permits the frozen previous hull as a competitor. Thus

$$A_j \leq A_{j-1} \leq \int_{\partial^* K_j} u_{j-1}^p \, \mathrm{d}\mathcal{H}^k_{g_0}. \tag{5}$$

These are hypotheses about actual hulls and velocities. A list of sampled points or a small finite residual does not establish them. The source supplies them on fixed time intervals for its uniform grid, with $\underline{u} = 4^{-T}$ and $\overline{u} = 1$.

Lemma 2.1 (One-sided frontier control)**.** *Let $a = \alpha/n$ and*

$$C_v = \max\{L c_*^{-a},\ B(c_* r_*^n)^{-a}\}. \tag{6}$$

Then $\sup_{\partial K_j} |v_{j-1}| \leq C_v (\Delta V_j)^a$. Only the filled-phase density at the new frontier is needed.

Proof. For $x \in \partial K_j$ let $\rho = \operatorname{dist}(x, K_{j-1})$. If $\rho = 0$, continuity and the zero value on the closed old conductor give $v_{j-1}(x) = 0$. For $0 < \rho < r_*$, every $r < \rho$ ball is disjoint from K_{j-1}, so $\Delta V_j \geq c_* r^n$. Let $r \uparrow \rho$ and use the Hölder bound relative to a nearest old-conductor point to obtain $|v_{j-1}(x)| \leq L(\Delta V_j / c_*)^a$. If $\rho \geq r_*$, use $r \uparrow r_*$ and $|v| \leq B$ instead. These alternatives also show that $\Delta V_j = 0$ forces every such value to vanish. There is no appeal to a graph through singular points or to a two-sided Hausdorff-distance estimate. □

Theorem 2.2 (Improved grid area loss)**.** *Under these hypotheses, with* $C_A = p\overline{u}^{p-1}\underline{u}^{-p}A_0 C_v$,

$$0 \leq A_0 - A_N \leq C_A \sum_{j=1}^{N} h_j (\Delta V_j)^a \tag{7}$$

$$\leq C_A \left(\sum_{j=1}^{N} h_j^{1/(1-a)} \right)^{1-a} V^a \leq C_A h_{\max}^a t_N^{1-a} V^a. \tag{8}$$

In particular, the source's uniform-grid estimate improves to $A_0 - A_j \leq C_T \epsilon^{\alpha/n}$ *for* $j\epsilon \leq T$.

Proof. Since $p > 1$, the mean-value inequality, positivity, and $v_{j-1} \leq 0$ give

$$0 \leq u_{j-1}^p - u_j^p \leq p\overline{u}^{p-1} h_j |v_{j-1}|.$$

Also $\mathcal{H}^k_{g_0}(\partial^* K_j) \leq \underline{u}^{-p} A_j \leq \underline{u}^{-p} A_0$. Apply (5) and Lemma 2.1, then sum the telescoping area differences. Hölder's inequality with exponents $1/(1-a)$ and $1/a$ gives the middle bound in (8). Finally $\sum h_j^{1/(1-a)} \leq h_{\max}^{a/(1-a)} t_N$. The source's finite-time support bound supplies a uniform upper bound for V, and its density and velocity bounds give mesh-independent C_A. Intermediate frozen frontiers have the same area as the corresponding grid frontier, so the conclusion covers the entire fixed time interval. □

2.1 Comparison, Sharpness, and Cost

The source `flow.tex`, lines 666–726, divides transitions according to whether $\Delta V_j > V_T \sqrt{\epsilon}$. It obtains $C_T(\sqrt{\epsilon} + \epsilon^{\alpha/(2n)})$. Theorem 2.2 retains its per-step comparison but sums all increments directly. The large-increment alternative in Lemma 2.1 is essential; applying a small-ball estimate beyond r_* would leave a gap.

For $T = V = 1$, $h_j = \Delta V_j = 1/N$, the sum in (7) is exactly N^{-a}, and both Hölder bounds are equalities. Thus the exponent a cannot be improved for arbitrary data satisfying only this scalar summation budget. This is not a lower bound for the error of an actual geometric flow. For instance, at $n = 3$, $\alpha = 1/2$, and $N = 64$, the normalized envelope is $1/2$, whereas the source's displayed two-term envelope is $1/8 + 1/\sqrt{2}$, before the possibly different constants. This comparison is symbolic, not a measured implementation speedup.

The sum retaining individual ΔV_j can be smaller than its uniform envelope, and gives a conditional adaptive error budget. It does not construct an adaptive hull solver or prove mesh-uniform constants for one. If a modified scheme instead has $\sup_{\partial K_j} |v_{j-1}| \leq C_v \Delta V_j^a + \zeta_j$, the same proof adds $p\overline{u}^{p-1}\underline{u}^{-p} A_0 \sum h_j \zeta_j$, provided all sign, freezing, positivity, and minimizing comparisons remain valid. A small energy residual by itself does not provide these pointwise ζ_j on a changing singular frontier.

3 A Sharp Mass-Defect Window

For the actual source flow, the mass evolution and estimates modules give

$$m' = -2\mu, \quad m \geq 0, \quad \mu \geq 0, \qquad b(t) := e^{2t}(m(t) - 2\mu(t)) \text{ is nondecreasing.} \tag{9}$$

Here m is locally absolutely continuous. For the theorem below take a right-continuous monotone representative of b and set $\mu = (m - e^{-2t}b)/2$. Equivalently, the distribution inequality is $D\mu + 3\mu\,\mathrm{d}t \leq m\,\mathrm{d}t$. This qualification excludes arbitrary changes of the value at one time, which distribution identities cannot detect.

Theorem 3.1 (Two-endpoint certificate)**.** *Suppose* (9) *holds on* $[s,t]$*, with the representative just specified. Put* $h = t - s > 0$*,* $z = e^h$*,* $M = m(s)$*,* $m = m(t)$*, and* $D = M - m$*. Then*

$$0 \leq \mu(t) \leq \mathcal{W}_h(M,m) := \frac{M - (2z - z^2)m}{2z(z-1)} = \frac{D}{2z(z-1)} + \frac{z-1}{2z}m. \tag{10}$$

This is sharp within the scalar differential-inequality class: equality holds throughout every window for $m(r) = Ae^{-r} + Be^{-2r}$ *and* $\mu(r) = Ae^{-r}/2 + Be^{-2r}$*,* $A, B \geq 0$*.*

Proof. Set $a = m - 2\mu$. Then $m' + m = a$ almost everywhere. Monotonicity in (9) implies $a(r) \leq e^{2(t-r)}a(t)$ for $s \leq r \leq t$. Integrate the exact identity $(e^r m(r))' = e^r a(r)$ backwards:

$$e^t m(t) - e^s m(s) \leq a(t)e^{2t}(e^{-s} - e^{-t}).$$

Dividing by e^s and substituting $a(t) = m(t) - 2\mu(t)$ gives $M \geq (2z - z^2)m + 2z(z-1)\mu(t)$, which is (10). For the displayed equality family, $a(r) = -Be^{-2r}$, so b is constant and every step of the proof is an equality. In particular no second distribution derivative or unproved differentiability of capacity is used. □

The equality family certifies scalar sharpness only; it is not asserted to be realizable by conformal flows with a nonempty horizon. Nonzero residual forcing in the differential inequality would require an additional weighted integral, not an unqualified use of (10).

Corollary 3.2 (Interval data and residual allowance)**.** *If* $0 \leq D \leq \overline{D}$ *and* $0 \leq m(t) \leq \overline{m}$*, then*

$$\mu(t) \leq \frac{\overline{D}}{2z(z-1)} + \frac{z-1}{2z}\overline{m}.$$

More generally, suppose $b = e^{2r}(m - 2\mu)$ *satisfies* $Db \geq -\rho(r)\,\mathrm{d}r$ *with* $\rho \geq 0$ *integrable on* $[s,t]$*, while* $m' = -2\mu$ *remains exact. Then add*

$$\frac{e^{-s}}{2z(z-1)} \int_s^t e^{-r} \int_r^t \rho(q)\,\mathrm{d}q\,\mathrm{d}r \tag{11}$$

to the right side of (10)*.*

Proof. The two coefficients in the D, m form are nonnegative. For the residual version, $b(r) \leq b(t) + \int_r^t \rho(q)\,\mathrm{d}q$. Substitute $a(r) = e^{-2r}b(r)$ in the integrating-factor proof. The added term before division by $2z(z-1)$ is precisely the numerator in (11). □

One may minimize these upper bounds over finitely many observed windows. Independent endpoint intervals can give $\overline{D} = \overline{m}(s) - \underline{m}(t)$, when consistent with an actual nonincreasing mass. The expression in M, m has a negative coefficient of m for $z < 2$, so replacing both endpoints blindly by upper bounds is not valid; the nonnegative D, m formulation avoids that mistake. For $h \downarrow 0$, the coefficients are $1/(2h) + O(1)$ and $h/2 + O(h^2)$. Small windows amplify endpoint uncertainty. Observed mass-loss information is indispensable: the theorem is not a universal convergence-time estimate.

4 Capacity on the Whole Actual Exterior

Fix one actual time slice with metric $h = g_t$, conductor $K = K_t$, and exterior E. Work with real functions and vector fields. Let $\mathcal{D}$ be the homogeneous energy completion of relative compactly supported smooth functions on the ambient manifold, allowing nonzero values on the buried physical boundary. The source's end Sobolev inequality fixes the zero-at-infinity realization. Quasicontinuous representatives define

$$\mathcal{V} = \{z \in \mathcal{D} : z = 0 \text{ quasi-everywhere on } K\}, \qquad \mathcal{A} = \{v \in \mathcal{D} : v = 1 \text{ quasi-everywhere on } K\}.$$

Here "quasi-everywhere" means except a set of zero Sobolev capacity, not merely zero perimeter measure. Assume, as in the source's capacity construction, that $\mathcal{A}$ is nonempty and closed and the energy norm is a Hilbert norm on $\mathcal{V}$. The minimizing potential W obeys

$$E_* := \int_E |\nabla W|_h^2 \, \mathrm{d}V_h = d\omega c, \qquad \int_E \langle \nabla W, \nabla z \rangle_h \, \mathrm{d}V_h = 0 \quad (z \in \mathcal{V}). \tag{12}$$

These are homogeneous-energy statements; no L^2 assumption on W is added.

Theorem 4.1 (Primal, dual, and imperfect-flux certificates)**.** *Let $v \in \mathcal{A}$, $q \in L^2(E, TE; h)$, and define*

$$E_v = \|\nabla v\|_2^2, \quad Q = \|q\|_2^2, \quad F = \int_E \langle q, \nabla v \rangle_h \, \mathrm{d}V_h, \quad D_q = \|\nabla v - q\|_2 .$$

All these quantities use the full exterior and the actual metric. If q is equilibrated, meaning $\int_E \langle q, \nabla z \rangle = 0$ for every $z \in \mathcal{V}$, and $Q > 0$, then

$$C_{\mathrm{low}} := \frac{F^2}{d\omega Q} \le c \le C_{\mathrm{up}} := \frac{E_v}{d\omega}, \qquad d\omega (C_{\mathrm{up}} - C_{\mathrm{low}}) = \|\nabla v - (F/Q) q\|_2^2 . \tag{13}$$

For an arbitrary q, if a certified number $\eta \ge 0$ satisfies

$$\sup_{z \in \mathcal{V},\ \|\nabla z\|_2 = 1} \left| \int_E \langle q, \nabla z \rangle_h \, \mathrm{d}V_h \right| \le \eta, \tag{14}$$

then

$$\max\left\{0, \frac{E_v - (D_q + \eta)^2}{d\omega}\right\} \le c \le \frac{E_v}{d\omega}. \tag{15}$$

Proof. Write $e = v - W \in \mathcal{V}$. Orthogonality in (12) gives the exact primal gap

$$E_v - E_* = \|\nabla e\|_2^2 . \tag{16}$$

If q is equilibrated, $F = \langle q, \nabla W \rangle_{L^2}$ is independent of the choice of $v \in \mathcal{A}$. Cauchy–Schwarz gives $F^2 \le Q E_*$. Expanding the square with the optimal scalar F/Q gives the gap identity in (13). More specifically, for any real λ,

$$\|\nabla v - \lambda q\|_2^2 = \|\nabla (v - W)\|_2^2 + \|\nabla W - \lambda q\|_2^2 ,$$

because both cross terms vanish against ∇e. This is the classical Prager–Synge orthogonality, not a newly discovered identity.

For the imperfect flux, let $r_v(z) = \int_E \langle \nabla v, \nabla z \rangle$ on $\mathcal{V}$. Equation (12) says its Riesz representative is e, so $\|r_v\|_{\mathcal{V}'} = \|\nabla e\|_2$. Splitting $\nabla v = (\nabla v - q) + q$ yields $\|r_v\|_{\mathcal{V}'} \le D_q + \eta$. Substitute this into (16), together with $E_* \ge 0$. □

If the inner boundary is smooth, q is weakly divergence free on all of E, and integration by parts with the full tail is justified, then

$$F = \int_{\partial E_{\text{inner}}} q \cdot n_E \, \mathrm{d}A_h. \tag{17}$$

The normal n_E points out of the exterior, into the conductor. The outer term $\int_{S_R} vq \cdot \nu$ must vanish as $R \to \infty$; this is an additional justification, not an automatic pointwise consequence of $q \in L^2$. For nonsmooth frontiers the weak pairing definition of F is preferable. It does not assign a classical normal to every singular point. The case $q = 0$ supplies only the trivial lower bound zero.

4.1 The Original Ram Comparison

Freshly read original TR8 [5], lines 65–197, treats the complete weak operator, trace lifting, conormal functional, and physical load measure. Its inverse is boundary dependent. Original TR250 [6], lines 457–594, develops the compatible primal/flux hypercircle mechanism. Theorem 4.1 applies that classical mechanism to the source's homogeneous, whole-exterior capacity, with its normalization $d\omega$ intact. Original TR6 [7], lines 506–615, proves state-dependent endpoint and transported-domain formulas under C^1 hypotheses. It expressly does not license a shape derivative across a singular minimizing frontier.

On regular compact patches, local Ram moments may represent a conforming v and a flux q, with physical mass and stiffness matrices computed from h. For a finite basis, the energy is the associated Gram quadratic form, not an unweighted coefficient norm. Interface normal-flux cancellation, zero-trace tests, and discarded-region terms must be verified before $\eta = 0$ or a small η can be asserted. An invertible panel matrix says nothing by itself about (14) on all of $\mathcal{V}$.

Small energy error also cannot be silently promoted to pointwise control of a boundary velocity. In a smooth $n \geq 3$ patch, a unit-height bump of radius r has squared gradient norm of order r^{n-2} while its supremum stays one. The density and Hölder controls used in Section 2 are therefore genuine extra information. Likewise, smoothly differentiating a regular patch neither follows a jumping hull nor proves detachment at a singular contact point.

5 An Exact Harmonic-Flat Tail Form

At a fixed normalized slice the source has $h = F^{4/d}\delta$ outside a sphere S_R, where $F > 0$ is Euclidean harmonic and tends to one. We now use F for this harmonic factor only; it is unrelated to the scalar flux pairing in (13). This subsection makes an omitted exterior energy computable from a whole boundary trace, when that trace and factor are actually known.

Proposition 5.1 (Physical transparent-boundary energy)**.** *Let $a \in H^{1/2}(S^{n-1})$ be real, and expand it in a real orthonormal spherical harmonic basis $Y_{\ell j}$ for the unit sphere: $a = \sum a_{\ell j} Y_{\ell j}$. Set*

$$f(r,\theta) = \sum_{\ell,j} a_{\ell j} (R/r)^{\ell+d} Y_{\ell j}(\theta), \qquad w = f/F.$$

Then w is the decaying h-harmonic extension of $a/F|_{S_R}$, and

$$\int_{r>R} |\nabla w|_h^2 \, \mathrm{d}V_h = R^d \sum_{\ell,j} (\ell + d) |a_{\ell j}|^2 + R^{n-1} \int_{S^{n-1}} a^2 \frac{\partial_r F}{F}(R,\theta) \, \mathrm{d}\theta. \tag{18}$$

Proof. Conformal powers give exactly $|\nabla w|_h^2 \, \mathrm{d}V_h = F^2 |\nabla w|_\delta^2 \, \mathrm{d}x$. Moreover $\operatorname{div}(F^2 \nabla w) = F\Delta f - f\Delta F = 0$. For a smooth finite harmonic trace, direct expansion and $\Delta F = 0$ give

$$F^2 |\nabla(f/F)|^2 = |\nabla f|^2 - \operatorname{div}(f^2 \nabla F/F).$$

At the inner boundary the outward normal of $\{r > R\}$ is $-\partial_r$; at infinity the last flux vanishes by the harmonic decay. Thus its contribution is the positive-sign boundary term in (18). The Dirichlet-to-Neumann value of the ℓth decaying mode is $(\ell + d)/R$, yielding the first term. Density of finite harmonics in $H^{1/2}$, boundedness of the smooth Robin multiplier, and continuity of the finite-energy extension give the asserted formula for general a. □

Although the second term may be negative, the total form is nonnegative because it is a physical energy. In general it couples different spherical harmonic modes. Dropping it is not the capacity of the source metric. For $a = a_L + e_L$, where a_L retains degrees at most L, orthogonality yields

$$\begin{aligned} |\mathcal{E}(a) - \mathcal{E}(a_L)| \le R^d \sum_{\ell > L, j} (\ell + d) |a_{\ell j}|^2 \\ + R^{n-1} \left\| \frac{\partial_r F}{F} \right\|_\infty (2 \|a_L\|_2 \|e_L\|_2 + \|e_L\|_2^2). \end{aligned} \tag{19}$$

Here $\mathcal{E}$ denotes the full form in (18); all trace norms are on the unit sphere. The right side requires a certified weighted high-mode tail, not just finitely many low moments. A single mode of degree $L+1$ may have small unweighted L^2 norm and a large energy after multiplication by $L+1+d$. An interior primal trial must match the chosen exterior trace exactly in $H^{1/2}$; otherwise it is not conforming. Uncertainty in the actual factor F needs a separate metric-error bound. This proposition does not infer that factor from a finite interpolation table.

5.1 Radial Worked Examples

Take the actual exterior $r \ge a_0 > 0$ with $h = (1 + br^{-d})^{4/d} \delta$, $b \ge 0$. The equilibrium potential and its normalization are

$$W(r) = \frac{a_0^d + b}{r^d + b}, \qquad c = a_0^d + b, \qquad m = 2b. \tag{20}$$

Indeed $(1 + br^{-d})W = (a_0^d + b) r^{-d}$ is Euclidean harmonic. Either its exterior energy or (18) gives $d\omega(a_0^d + b)$. The two terms in (18) are respectively $d\omega(a_0^d + b)^2/a_0^d$ and $-d\omega b(a_0^d + b)/a_0^d$. This explicitly verifies the sign and the metric correction. At the Schwarzschild horizon $a_0^d = b$, $c = m = 2b$, and the boundary area is $\omega(4b)^{k/d}$, consistent with (2). For $b = 0$, the Euclidean exterior ball has $c = a_0^d > 0$ but $m = 0$. Its normal-toward-the-end mean curvature is positive, so it does not satisfy the source's mass-capacity comparison hypotheses.

For a completely explicit certificate in this Euclidean exterior, write $x = (a_0/r)^d$, $W = x$, and $v = x + \theta(x - x^2)$ for any real θ. The trace is exactly one and the infinity value exactly zero. Direct radial integration gives

$$\frac{\|\nabla(x - x^2)\|_2^2}{d\omega} = a_0^d/3, \qquad C_{\mathrm{up}} = a_0^d(1 + \theta^2/3). \tag{21}$$

The flux $q = \nabla W$ is equilibrated on the whole exterior; it gives $C_{\mathrm{low}} = a_0^d$ and an exact gap $a_0^d \theta^2/3$. This tests the normalization and classical certificate, not a nonspherical flow, singular geometry, or the global inequality.

6 Combining the Certificates

The three mechanisms address different uncertainties. Section 2 controls finite-grid area loss from actual swept volume. Section 4 controls a capacity calculation on one specified actual slice. Section 3 controls the instantaneous defect from actual mass endpoints. They cannot be substituted for one another.

Corollary 6.1 (Two independent upper bounds)**.** *On an actual flow slice with $m' = -2(m - c)$ and $\mu = m - c \geq 0$, assume a valid capacity lower bound C_{low} and exact endpoint masses. For every available $h \in (0, t]$,*

$$0 \leq \mu(t) \leq \min\{m(t) - C_{\mathrm{low}},\ \mathcal{W}_h(m(t-h), m(t))\}.$$

With uncertain data use $\overline{m}(t) - C_{\mathrm{low}}$ and the nonnegative coefficient formula of Corollary 3.2 instead.

Proof. The first bound is the definition of μ and $c \geq C_{\mathrm{low}}$; the second is Theorem 3.1. A negative resulting upper bound indicates inconsistent certificates, not a license to set the true defect to a negative number. □

There is also an exact capacity-to-mass transfer along the same flow:

$$m(t) = e^{-2(t-s)} m(s) + 2 \int_s^t e^{-2(t-r)} c(r)\, \mathrm{d}r. \tag{22}$$

This follows by multiplying $m' + 2m = 2c$ by e^{2t}. Thus measurable whole-flow bounds $c_- \leq c \leq c_+$ give the corresponding convolution brackets. If the capacity error is at most ε at every intervening time, its mass contribution is at most $\varepsilon(1 - e^{-2(t-s)})$; an initial mass error is multiplied by $e^{-2(t-s)}$. Finitely many time samples do not verify the intervening inequality. This stable scalar transfer cannot supply a missing hull, mass-capacity inequality, or spatial derivative estimate.

6.1 Why No Global Stopping Time Follows

Even in the scalar class (9), a mass loss can be delayed arbitrarily. Fix $L > 0$. Set $m(t) = 1$, $\mu(t) = 0$ for $t \leq L$, and for $\tau = t - L \geq 0$ set

$$m(t) = 2e^{-\tau} - e^{-2\tau}, \qquad \mu(t) = e^{-\tau} - e^{-2\tau}.$$

Both are nonnegative, $m' = -2\mu$, and $e^{2t}(m - 2\mu)$ increases up to L and is constant thereafter. Yet $\mu(L + \log 2) = 1/4$ for every L. This is a scalar counterexample to a uniform decay time derived only from $m(0) = 1$ and (9). It is not asserted to be a geometric flow. The source's qualitative argument that $\mu \to 0$ is unaffected.

7 Current Literature and Proof Boundaries

The public refresh on October 9, 2026 inspected primary metadata and selected main statements, not complete external proofs. Bray–Lee's published 2009 work establishes the dimension-below-eight baseline [2]. Its primary arXiv Theorem 1.4 states the numerical bound for a one-ended exterior with a full outer-minimizing horizon; the source notes the published numerical theorem separately from rigidity qualifications. We make no new rigidity claim.

Bi–Zhu's current v2, dated September 15, 2026, claims the all-dimensional inequality and equality characterization [3]. The HTML Theorem 1.2 explicitly includes an almost-minimizing Caccioppoli boundary, an outermost minimizing-enclosure condition, minimal regular part, and a $C^{2,\alpha}$ metric

extension. These details should not be replaced by an abstract slogan about arbitrary singular boundaries. Our assigned source uses stronger local minimizing and smooth ambient hypotheses, and reconstructs its own numerical argument. Its proof is not independently certified by this report.

Brendle–Wang v2, dated April 20, 2026, is a separate positive-mass input [4]. Its Definition 1.1 uses all-order asymptotic decay and Corollary 1.6 assumes scalar curvature positive at every point. The assigned source's repair and completion steps address the passage to the metric class needed in its argument. We do not replace those steps by the claim that any local harmonic factor with a small monopole is already a complete positive-mass comparison metric. All three primary sources remain credited to their authors.

8 Advantages, Costs, and Further Work

The direct mesh sum is genuinely stronger than the source's displayed large/small-transition bound under the same fixed-time inputs. It treats singular frontier points through density, uses only one-sided swept volume, and admits nonuniform budgets. Its constants include the actual density scale, Hölder modulus, and lower conformal bound; these may deteriorate with T. It gives no quantitative convergence rate for the entire metric or moving hull, and no long-time normalized convergence modulus.

The mass-window certificate is exact in the coupled scalar class and can use an observed window without a pointwise capacity calculation. It amplifies endpoint errors at short windows and still requires actual flow masses. The dual capacity certificate can be sharp with an appropriately scaled equilibrated flux, but constructing such a flux on the whole exterior may be harder than constructing the primal potential. A small local polynomial residual alone does not certify it. The exact tail form is useful when a verified harmonic-flat factor and trace tail are available, not when only a finite interpolant is known.

Promising conditional extensions are adaptive grids with proved uniform static-hull constants, equilibrated flux reconstruction with explicit interface and infinite-tail estimates, and interval mass observations synchronized with the actual flow. Each needs its missing geometric or analytic hypotheses. No uniform regular chart through a singular set, global finite-dimensional inverse, or effective positive-mass modulus has been supplied here. The improvement is quantitative and local to the assigned proof mechanisms; it is not a new proof of the Penrose inequality, an equality theorem, or an empirically demonstrated computational speedup.

9 Read Scope and Reproducibility

The owned ledger records pre-read and post-read SHA256 values, literal display intervals, and their exact unread complements. Entire fresh reads include the introduction, flow, mass evolution, estimates, geometry statements, median-normalization figure, and end-limit module, as well as the root driver, preamble, and README. Partial fresh reads include mass-capacity lines 1–165, regularity-separation 1–180, completion 1–150, approximation 1–110, and references 23–89. The enclosure proofs and weighted-minimizer module were not literally read. Their use above is explicitly imported; mapping their position in the architecture does not count as proof inspection.

The original private proof intervals are TR8 lines 65–197, TR250 lines 457–594, and TR6 lines 506–615. They remain local and confidential. No previous comparative report, parent's checks, or whole-corpus read is counted as the author's evidence. Captured-tree identity against the October 8 snapshot is provenance only, not current branch status or mathematical validity. The fresh scope

totals 2,683 of 5,616 public text lines and 381 private-original lines; all remaining intervals are explicitly listed as unread.

The companion owned script checks exact scalar mesh budgets, the two-exponential window and residual identities, capacity projection identities, and radial normalizations: 2,004 exact-rational cases and 4,913 assertions passed. Its final JSON distinguishes parameter cases from assertions; no numerical, parent, or foreign-checker cases are included. These fixtures supplement the written proofs; they do not verify global existence, regularity, detachment, positive mass, or asymptotic geometry. The final freeze records both deliverables, owned evidence, test code and results, build records, all final page images, and read-only source endpoints.

References

[1] Assigned public manuscript, *Conformal flow and the Riemannian Penrose inequality with minimizing frontiers*, September 27, 2026, captured OpenAI math preprint tree. Local driver `build/paper.tex`; scientific modules and exact hashes are listed in the owned read ledger. https://github.com/openai/math.

[2] H. L. Bray and D. A. Lee, *On the Riemannian Penrose inequality in dimensions less than eight*, Duke Math. J. 148 (2009), 81–106. https://doi.org/10.1215/00127094-2009-020; primary preprint https://arxiv.org/abs/0705.1128.

[3] Y. Bi and J. Zhu, *Riemannian Penrose inequality in all dimensions*, arXiv:2605.00680v2, September 15, 2026. https://arxiv.org/abs/2605.00680v2. Main statement inspected; full proof not independently audited.

[4] S. Brendle and Y. Wang, *A dimension descent scheme for the positive mass theorem in arbitrary dimension*, arXiv:2604.08473v2, April 20, 2026. https://arxiv.org/abs/2604.08473v2. Definitions and main corollary inspected; full proof not independently audited.

[5] M. Subbarao, original private Ram PDE report TR8, `RamNonlin-IE-PDE-astra/TR8-lin-non-lin/TR8-lin-non-lin.tex`, lines 65–197: weak domains, boundary liftings, conormal functionals, and physical Green normalization. Confidential local manuscript.

[6] M. Subbarao, original private Ram foundation report TR250, `RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`, lines 457–594: compatible primal and equilibrated-flux hypercircle identities. Confidential local manuscript; classical orthogonality credited.

[7] M. Subbarao, original private nonlinear PDE report TR6, `RamNonlin-IE-PDE-astra/TR6-lin-non-lin/TR6-lin-non-lin.tex`, lines 506–615: state-dependent supports and transported-domain derivatives under explicit smoothness hypotheses. Confidential local manuscript.